

Project Planning Proposal: Customer Lifetime Value Modeling for Online Retail

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Abstract

This proposal describes a four-week Data Science project for ADY201m. The project predicts Customer Lifetime Value (CLV) from transactional data of a UK online retailer and ranks customers to focus marketing spend on high-value (VIP) segments. Because future CLV is zero-inflated, the modelling plan includes a two-stage LightGBM Hurdle model that separates return prediction from positive revenue estimation. Following the mandated PBL + RBL framework, the plan covers a full pipeline (ingestion, cleaning, EDA, feature engineering, modelling) in Python with supporting SQL analytics, at least five regression models compared against a published baseline, an interactive dashboard, and a complete AI Audit Log demonstrating the *Human Delta*.

Keywords: Customer Lifetime Value, RFM, Hurdle Model, BG/NBD, Marketing Analytics, Online Retail, SQL, Python

1 Project Name and Requirements

Project name: *Customer Lifetime Value Modeling for Online Retail – Prioritising Marketing Spend on VIP Customers via Predicted CLV.*

ADY201m course requirements addressed: a complete pipeline (Data → Cleaning → EDA → Features → Modelling → Evaluation); at least five ML models compared against a published baseline; SQL analytics integrated with Python; an interactive dashboard (Power BI; Plotly Dash fallback); a full AI Audit Log (15–20 core prompts and ≥ 3 hallucinations with Human Delta); a 10–12 page Final Report

typeset in the Springer Nature single-column LaTeX template; and an IBM Coursera certificate for the +1 bonus.

2 Business Understanding and Analytic Approach

E-commerce marketing budgets are typically spread evenly across the customer base, yet retail revenue is highly concentrated – empirically, the top 10% of customers generate 60–80% of revenue. Spending retention dollars on the long tail yields low return on investment. The business problem is therefore to *identify the customers most likely to deliver the highest future revenue* so that retention and cross-sell campaigns can be concentrated on them.

The analytic approach is structured as four progressively prescriptive layers: *Descriptive* (RFM and cohort retention), *Diagnostic* (SQL RFM segmentation via Q7; statistical tests on CLV between segments), *Predictive* (five-plus regression models forecasting 6-month CLV), and *Prescriptive* (a dashboard ranking the top-10% VIPs for marketing activation).

The roles of RFM, CLV, and Retention. The three concepts in this study are *complementary, not interchangeable*: (i) **RFM** acts as a *descriptive segmentation lens*, grouping customers by aggregate Recency–Frequency–Monetary behaviour into five named clusters (Champion, Loyal, At-risk, Lost, New) used both as model input features and as a transparent comparison baseline; (ii) **CLV** acts as the *predictive ranking target*, a supervised regression label equal to the per-customer revenue in the future prediction window, used to rank customers by expected future value; (iii) **Retention** acts as the *validation lens*, verifying via a held-out temporal window whether customers ranked high by predicted CLV actually return and contribute revenue, thereby confirming that predicted CLV reflects long-term worth rather than a short-term monetary spike. RQ1 exercises RFM, RQ2 exercises CLV ranking, and RQ3 exercises retention validation; together they close the loop from segmentation to prediction to business validation.

3 Research Questions (RBL)

- **RQ1.** Which RFM segment has the highest mean CLV, and are the differences statistically significant? *Method:* SQL quintile RFM segmentation (Q7) + ANOVA / Kruskal–Wallis on mean CLV by segment.
- **RQ2.** Which model predicts CLV most accurately relative to the BG/NBD + Gamma-Gamma baseline? *Method:* train ≥ 5 models; report MAE, RMSE, Normalized Gini, Revenue Capture@10%; paired *t*-test across walk-forward folds.
- **RQ3.** Do customers with high predicted CLV actually retain at a higher rate in a held-out period? *Method:* 6-month temporal holdout, Precision@Top-10%, Lift over random targeting.

4 Data Collection, Understanding, and Preparation

Source. *Online Retail II* (UCI ML Repository / Kaggle): $\sim 1.07\text{M}$ transactions, $> 5,800$ unique customers, 4,631 stock codes, covering 1 Dec 2009 to 9 Dec 2011 for a UK gift retailer. Key fields are `InvoiceNo`, `StockCode`, `Quantity`, `InvoiceDate`, `Price`, `CustomerID` and `Country`.

Preparation pipeline. (i) merge the two yearly sheets and parse `InvoiceDate`; (ii) drop rows with missing `CustomerID`, non-positive quantity or price (cancellations are handled separately); (iii) derive `TotalPrice = Quantity  $\times$  Price`; (iv) apply an 18-month observation window (Dec 2009 – May 2011) and a 6-month prediction window (Jun 2011 – Dec 2011); (v) compute RFM and behavioural features (`Tenure`, `ActiveMonths`, `AvgOrderValue`, `ProductDiversity`, `CancellationRatio`, `IsUK`) on the observation window; (vi) compute the supervised label `ActualCLV` as the total revenue per customer in the prediction window.

5 Proposed Methodology and Tools

This project follows a complete end-to-end data science pipeline (ingestion $\rightarrow$ cleaning $\rightarrow$ EDA $\rightarrow$ feature engineering $\rightarrow$ modelling $\rightarrow$ evaluation $\rightarrow$ deployment). The methodology integrates: (i) **SQL analytics** on DuckDB for efficient RFM, cohort, and segmentation queries; and (ii) **Python machine learning** for feature engineering, predictive modelling, and evaluation. Key tools: `pandas`, `scikit-learn`, `LightGBM`, `lifetimes`, `SHAP`, `Optuna`, `Plotly`; **SQL**: DuckDB for **seven** analytical queries (Q1–Q7) supporting the three RQs; **Dashboard**: Plotly Dash (Power BI fallback). Hyperparameter tuning uses Optuna under walk-forward cross-validation; headline metrics are Normalized Gini and Revenue Capture@10%.

6 Data Analysis with SQL

Seven analytical queries on DuckDB directly support the three RQs:

- `q1_rfm_per_customer` — R/F/M aggregation per customer (RQ1);
- `q7_rfm_segmentation` — quintile *R/F/M* scores and named RFM groups (Champion, Loyal, At-risk, Lost, New); satisfies Project 8 requirement for RFM group segmentation (RQ1);
- `q2_cohort_retention` — first-purchase-month $\times$ observation-month retention matrix (RQ2/RQ3);
- `q3_top_vip_by_monetary` — top-100 Monetary customers (RQ3);
- `q4_revenue_by_country` — UK vs. non-UK breakdown (RQ1);
- `q5_churn_candidates` — Recency > 90 days, tiered by Monetary (RQ3);
- `q6_monthly_active_customers` — monthly active counts and revenue (RQ2).

Deliverables: self-contained `02_sql_analysis.ipynb`; seven CSV files in `notebooks/outputs/` (`q1_rfm.csv`–`q7_rfm_segments.csv`); one-paragraph summary linking each query to an RQ. SQL source files live in `notebooks/sql/`.

7 Data Analysis with Python

Five reproducible Python modules: `01_data_cleaning.py` (ingestion / cleaning), `02_eda.py` (descriptive statistics and missing-value patterns), `03_feature_engineering.py` (RFM, behavioural, and monthly-sequence features: 18 months $\times$ 3 channels – revenue, frequency, average order value), `04_clv_labels.py` (compute ActualCLV on the prediction window), `05_segmentation.py` (optional K -means comparison; primary named segments come from SQL Q7 and feed the dashboard). Required libraries: `pandas`, `numpy`, `scikit-learn`, `xgboost`, `lightgbm`, `lifetimes`, `shap`, `matplotlib`, `seaborn`, `plotly`.

8 Data Visualisation

Six figures support the report and the dashboard: (1) RFM scatter / pair plot with K -means clusters; (2) cohort retention heat-map; (3) violin plot of CLV per segment; (4) Pareto chart (80/20 cumulative revenue check); (5) Revenue Capture curves comparing all trained models; (6) Predicted vs. Actual CLV scatter (calibration). All figures are exported at 300 dpi PNG.

9 Regression Analysis ($\geq$ Five Models)

Why a Hurdle architecture is needed. Exploratory analysis on Online Retail II shows that 47–67% of active customers in the observation window generate *zero revenue* during the prediction window, while non-zero customers exhibit a heavy-tailed spending distribution (skewness > 22). A single MSE-trained regressor (Linear, Random Forest, or XGBoost) is therefore optimised against two conflicting objectives at once: (i) pulling predictions toward zero for the large no-purchase mass, and (ii) tracking the long-tail VIP customers. The Hurdle architecture decouples these: one classifier learns *whether* the customer returns, one regressor learns *how much* a returner spends, and predictions are recombined multiplicatively. This is the single methodological choice the project claims as a contribution; the four other models in Table 1 are deliberate ablations of components the Hurdle subsumes.

Evaluation metrics (grouped by purpose). To avoid an indiscriminate metric panel, the evaluation is organised into three purpose-specific tiers, each linked to the research question it serves:

- **Primary — ranking quality** (drives RQ2, RQ3, and the final model selection): **Normalized Gini** (NG) and **Revenue Capture@10%** (RC@10%). NG measures how well the model orders customers by future revenue; RC@10% measures how much actual revenue is captured by contacting only the top-10% predicted VIPs. Both directly map to the marketing-budget allocation use case.
- **Secondary — regression quality** (calibration sanity checks for profit forecasting): **MAE** and **RMSE** on the dollar scale. Reported but not used for headline selection, because zero-inflation makes raw error metrics misleading on this label.

Table 1 Five regression models (plus probabilistic baseline) spanning four families. Each model is selected to address a specific weakness of the model above it; the LightGBM Hurdle is the proposed contribution and explicitly handles the 47–67% zero-inflation that single-loss regressors cannot.

Model	Family	Loss	Rationale and what it adds
BG/NBD + GG	Probabilistic	— (MLE)	Published baseline (Fader et al., 2005) using only RFM summaries under permanent-churn and frequency–value-independence assumptions; defines the bar all learned models must beat.
Linear / Ridge	Linear	MSE on $\log(1+y)$	Interpretable linear baseline; quantifies how much CLV is explained by additive RFM-and-behavioural effects before any non-linearity.
Random Forest	Bagging tree	MSE	Captures non-linear interactions among RFM and behavioural features; robust to outliers via averaging, but underweights heavy-tailed VIPs.
XGBoost	Boosting tree	MSE on raw / log target	Strongest single-stage tabular boosting baseline; serves as the “single-loss” upper bound that still treats zero and positive customers under one objective.
LightGBM Hurdle (<i>proposed</i>)	Two-stage tree	BCE + MSE on $\log(1+y)$	Decomposes the zero-inflated problem into Stage 1 : $P(y>0 \mid x)$ (LightGBM classifier on all customers), and Stage 2 : $\mathbb{E}[\log(1+y) \mid y>0, x]$ (LightGBM regressor on the positive subset). Predictions are combined via $\hat{y} = \hat{\pi} (e^{\hat{m}} - 1) e^{\hat{\sigma}^2/2}$, addressing the limitation that single-loss regressors cannot disentangle <i>whether</i> a customer returns from <i>how much</i> they spend.

- **Diagnostic** (stability and qualitative interpretation, not headline numbers): R^2 on the positive subset only, **Spearman** rank correlation (rank stability against MAE), and **Precision@Top-10%** (binary VIP hit rate, supports RQ3’s retention claim).

We deliberately drop *Lift@10%* from the headline panel because $\text{Lift@10\%} = \text{RC@10\%}/10\%$ is a linear rescaling of RC@10% and adds no independent information. Pairwise statistical comparison against the BG/NBD + Gamma-Gamma baseline is conducted on the *primary* tier only, using paired t -tests across walk-forward folds with bootstrap 95% CIs.

10 Data Analysis with a Tool (Dashboard)

A four-page Power BI dashboard (Plotly Dash as a fallback): (1) *Overview* – KPIs (revenue, customers, retention, AOV); (2) *RFM Matrix* – interactive heat-map with customer drill-down; (3) *CLV Forecast* – top-100 predicted VIPs with confidence intervals; (4) *Campaign Targeting* – segment filter and CSV export for the marketing team.

11 Four-Week Timeline and Deliverables

Table 2 Four-week schedule with weekly deliverables and deadlines.

Week	Activities and Deliverables	Deadline
W1	Repository & env. setup; data cleaning and EDA; baseline RFM; research questions locked. <i>Out:</i> <code>customers.csv</code> , EDA notebook, dataset statistics table.	End of W1
W2	Seven SQL queries (Q1–Q7) answering RQ1–RQ3; full RFM, behavioural and monthly-sequence features; compute <code>ActualCLV</code> . <i>Out:</i> <code>02_sql_analysis.ipynb</code> , <code>q1_rfm.csv</code> – <code>q7_rfm.segments.csv</code> , <code>features.csv</code> , cohort heat-map, Pareto chart.	End of W2
W3	Train and tune five regression models under walk-forward CV; compute the full metric panel; SHAP on the best model. <i>Out:</i> <code>metrics_comparison.csv</code> , six figures, SHAP report.	End of W3
W4	Build the four-page Power BI dashboard; write the 10–12 page Springer LaTeX final report; finalise the AI Audit Log (15–20 prompts; ≥ 3 hallucinations). <i>Out:</i> <code>dashboard.pbix</code> , <code>final_report.pdf</code> , <code>AI_AuditLog_Template.xlsx</code> .	End of W4

Final deliverables. (i) This Project Planning PDF; (ii) source code on GitHub with a single-command reproduction script; (iii) the Springer-formatted Final Report; (iv) the dashboard; (v) the AI Audit Log workbook; (vi) the IBM Coursera certificate link.

12 Risks, Mitigations, and AI Reflection

Risks. (i) $\sim 25\%$ of rows lack `CustomerID` – these are filtered and the impact is documented; (ii) the label distribution is heavily zero-inflated, which causes negative R^2 for naive regressors – mitigated by the Hurdle two-stage architecture; (iii) Power BI installation may fail on some machines – a Plotly Dash fallback is pre-prepared; (iv) AI hallucination is mitigated by mandatory cross-checking and full disclosure in the Audit Log.

AI Reflection. Every Generative AI interaction is recorded with (a) the original prompt, (b) the AI response, (c) the manual edits applied, and (d) the Human Delta – the domain judgement, validation or correction added beyond the AI output. The proposal is aligned with the nine criteria of the Final Project Evaluation rubric and the 30% weight reserved for AI Audit & Reflection.